Thermodynamic Ledger Stabilization and Trophic Cascade Refinement in Spatial Ecological Networks: Sprint 028 Report

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Sprint 028 Technical Report

Abstract

Ecosystem simulation frameworks frequently struggle to reconcile localized metabolic activity with global conservation laws. In this preprint, we summarize the architectural advancements introduced in **Sprint 028** of the Web of Life project. We formalize a rigorous thermodynamic ledger framework that enforces the First Law of Thermodynamics (mass-energy conservation) across spatial graphs with sub-atomic precision ($\Delta M < 10^{-12}$) alongside the Second Law via explicit entropy generation ($\Delta S = Q/T$). Furthermore, we introduce spatial nutrient gradients (`SpatialNode`, `BiomePatch`) and expand our trophic hierarchy to include specialized detritivore monad stock transitions, ensuring zero matter loss during organismal senescence and scavenging.

1 Introduction and Theoretical Framework

Traditional ecological modeling often treats nutrient pools and organismal biomass as loosely coupled compartments, frequently suffering from numerical drift or unphysical mass creation/destruction. Within the Web of Life paradigm, organisms and environmental patches are unified under a rigorous thermodynamic and mass-balanced monad architecture.

Sprint 028 bridges spatial resource distribution, organismal movement costs, and multi-tier trophic cascades. The central objectives are:

1. **Strict Mass Conservation:** Guaranteeing that atomic stocks (C, N, P, H_2O) are strictly conserved across all state transitions.
2. **Exergy Dissipation & Entropy Tracking:** Quantifying metabolic and kinetic heat loss to increment global entropy monotonically.
3. **Spatial Equilibrium:** Implementing continuous diffusion-consumption nutrient patches (`BiomePatch`) that govern autotrophic uptake and heterotrophic foraging.

Official repository reference: `https://github.com/pascalranoroarijaona/WebOfLife`.

2 Core Mathematical Formalization

2.1 First and Second Laws of Thermodynamics

For any closed system simulation graph, the total mass remains invariant:

$$\sum \text{Mass}_{\text{inputs}} = \sum \text{Mass}_{\text{outputs}} + \text{Mass}_{\text{excreted/resorbed}} \tag{1}$$

Metabolic transformations incur obligatory heat dissipation (Q). The global entropy increment ΔS_{total} satisfies:

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \frac{Q_{\text{dissipated}}}{T_{\text{ambient}}} \geq 0 \tag{2}$$

2.2 Detritivore Scavenging & Decomposition

Upon organismal death, biomass is converted into a `Carcass` and deposited directly into the local `BiomePatch`. Detritivores process this carcass with an assimilation efficiency $\eta_{\text{detritivore}} \approx 0.15$:

$$\text{Assimilated} = \eta \cdot \text{Carcass}, \quad \text{Residue} = (1 - \eta) \cdot \text{Carcass} \tag{3}$$

Oxidation of unassimilated carbon releases thermal energy $Q = C_{\text{carcass}} \times 10.5$ J, directly updating the `ThermodynamicLedger`.

3 System Architecture & Implementation

The implementation extends our core abstract classes without introducing parallel inheritance trees:

- `SpatialNode` / `BiomePatch`: Manages localized carrying capacities and elemental stocks.
- `ThermodynamicLedger`: Audits mass conservation and tracks cumulative system entropy.
- `DetritivoreMonad`: Executes circular matter transitions, preventing ecological deadlocks.

```
@dataclass
class ThermodynamicLedger:
    total_entropy: float = 0.0
    total_dissipated_heat: float = 0.0
    initial_system_mass: Optional[ElementalStocks] = None

    def record_dissipation(self, joules: float, ambient_temp: float =
        298.15) -> None:
        if joules < 0:
            raise ValueError("Dissipated heat cannot be negative.")
        self.total_entropy += joules / ambient_temp
        self.total_dissipated_heat += joules

    def audit_mass_conservation(self, current_mass: ElementalStocks) ->
        float:
        if self.initial_system_mass is None:
            self.initial_system_mass = current_mass
            return 0.0
        return float(
            abs(current_mass.carbon - self.initial_system_mass.carbon)
                +
            abs(current_mass.nitrogen - self.initial_system_mass.
                nitrogen) +
            abs(current_mass.phosphorus - self.initial_system_mass.
                phosphorus) +
            abs(current_mass.water - self.initial_system_mass.water)
        )
```

4 Verification and Acceptance Criteria

Sprint 028 establishes rigorous automated verification suites:

1. **Conservation Test:** Over 1,000 simulation steps, total atomic mass discrepancy must not exceed 10^{-12}.
2. **Entropy Growth Test:** $\sum \Delta S$ is verified to be monotonically non-decreasing.
3. **Trophic Balance:** Phase portraits of predator-prey oscillations exhibit stable limit cycles without numerical collapse.

5 Conclusion

Sprint 028 successfully establishes a closed-loop thermodynamic ecosystem engine within the Web of Life framework. By enforcing strict elemental accounting and explicit entropy generation, we pave the way for long-term evolutionary simulations grounded in fundamental physical laws.